4.1 $\hat{y} = b_0 + \sum_{k=1} b_k x_k$, $\frac{d\hat{y}}{dx_k} = b_k$

For a 1 unit increase in x_k, the model changes the prediction by b_k.

4.2 Poop poop poop UwU

$\hat{p} = \frac{e^{b \cdot x}}{1+e^{b \cdot x}}$; u sub where $U = b \cdot x$ | $\hat{p} = \frac{e^u}{1+e^u} \rightarrow \frac{dp}{du} = \hat{p}(1-\hat{p})$

$\hookrightarrow \frac{d\hat{p}}{dx_k} = \frac{d\hat{p}}{du} \cdot \frac{du}{dx_k} = b_k \cdot \hat{p}(1-\hat{p})$

As x changes, my answer changes. It is similar with having a b_k but it just depends on $\hat{p}$ instead of b_k being alone.

Try $p=0 \rightarrow 0(1-p) = 0 \cdot 1 = 0$

Try $p=1 \rightarrow 1(1-1) = 0$

Try $p=0.5 \rightarrow 0.5(1-0.5) = 0.25$

Values where $p \simeq 0.5$ b/c 0.25 is the max value is best for good estimate of how a change in x_k affects the prediction

4.3

It changes by having One unit increase in x_k increases the log odds ratio by b_k.